



## Opinion piece



**Cite this article:** Göpel T. 2026 How should we address non-reproducibility? *Phil. Trans. R. Soc. B* **381**: 20250057.

<https://doi.org/10.1098/rstb.2025.0057>

Received: 25 April 2025

Accepted: 6 September 2025

One contribution of 20 to a theme issue

'Embracing variability in comparative physiology: why it matters and what to do with it'

### Subject Areas:

physiology, theoretical biology

### Keywords:

reproducibility crisis, reporting guidelines, open science, publishing policy, scientific rigour

### Author for correspondence:

Torben Göpel

e-mail: [torben.goepel@uni-goettingen.de](mailto:torben.goepel@uni-goettingen.de)

# How should we address non-reproducibility?

Torben Göpel<sup>1,2</sup>

<sup>1</sup>Multiscale Biology, Johann-Friedrich-Blumenbach-Institut für Zoologie und Anthropologie, Georg-August-Universität Göttingen, Göttingen, Niedersachsen, Germany

<sup>2</sup>Department of Biological Sciences, University of North Texas, Denton, TX, USA

TG, 0000-0002-6726-6319

Inter-experimental variability poses a significant challenge to reproducibility in experimental biology, hindering scientific progress and the translation of research findings. This variability arises from numerous factors, including subtle differences in reagents, equipment calibration, environmental conditions and unaccounted biological variation. Despite the existence of guidelines such as ARRIVE (Animal Research: Reporting of In Vivo Experiments) and MDAR (Materials Design Analysis Reporting), the reporting of crucial experimental details often remains incomplete or inconsistent across studies. Recent analyses have suggested that the insufficient reporting of materials and methods may substantially contribute to non-reproducibility in biological research. This underscores the need for more comprehensive and standardized reporting practices as the best approach to ensure good science. To address this issue, publishers, editors and reviewers must enforce existing standards more rigorously, while authors and researchers should prioritize the comprehensive reporting of experimental details. Innovative approaches such as experimental heterogenization and open science initiatives are emerging to enhance reproducibility. A multifaceted approach involving stricter guideline enforcement, recognition of replication studies and fostering a culture that values reproducibility over publication speed and sensationalization is essential for improving the reliability and credibility of experimental biology research.

This article is part of the theme issue 'Embracing variability in comparative physiology: why it matters and what to do with it'.

## 1. Causes of non-reproducibility

Inter-experimental variability is a key contributor to non-reproducibility in experimental biology. Factors (figure 1) leading to inter-experimental variability include subtle variations in reagents [1,2], differences in equipment and its adjustment/calibration [3], environmental conditions [4] and unaccounted biological differences within experimental subjects (e.g. sampling bias and cryptic variability) [5]. Differences in one or more of these experimental aspects can lead to discrepancies in the results when the experiments are repeated, undermining the reproducibility of the findings [6]. Consequently, this variability not only affects the reliability of individual studies but also impacts the broader scientific community's ability to build upon and validate previous research, ultimately hindering scientific progress and the translation of findings into practical applications.

In a survey of 1576 scientists, not only did 90% of the scientists agree that there is a reproducibility crisis in science, but insufficient reporting<sup>1</sup> was also

<sup>1</sup>In Baker [7], reporting issues were referred to as 'selective reporting' whereas I prefer to remain agnostic about whether insufficient reporting happened intentionally or not.

given as the number 1 factor leading to non-reproducibility [7]. Non-reproducible results often raise suspicion of misconduct or fraud, and cases of young upcoming researchers who have shaped not only their careers but also their entire field based on fabricated data are well known. Although scientific misconduct may not be the sole or even main reason for non-reproducibility [7–10], the fact that significant funding goes towards irreproducible, often fabricated science is alarming [11].

There are different levels of causality to non-reproducibility that are connected to each other. This is exemplified by a hypothetical yet realistic sequence of events: the pressure to publish [7,12,13] may motivate some researchers to engage in data dredging via *post hoc* removal of data to achieve (erroneously) statistically significant results. Subsequently, insufficient oversight by the respective principal investigator [7] can lead to selective reporting of data and the analysis thereof in a manuscript, which then may be insufficiently reviewed [7], ultimately leading to the publication of bad science. Then, a grant application based on these fabricated results is funded, unknowingly rewarding scientific misconduct [11,12].

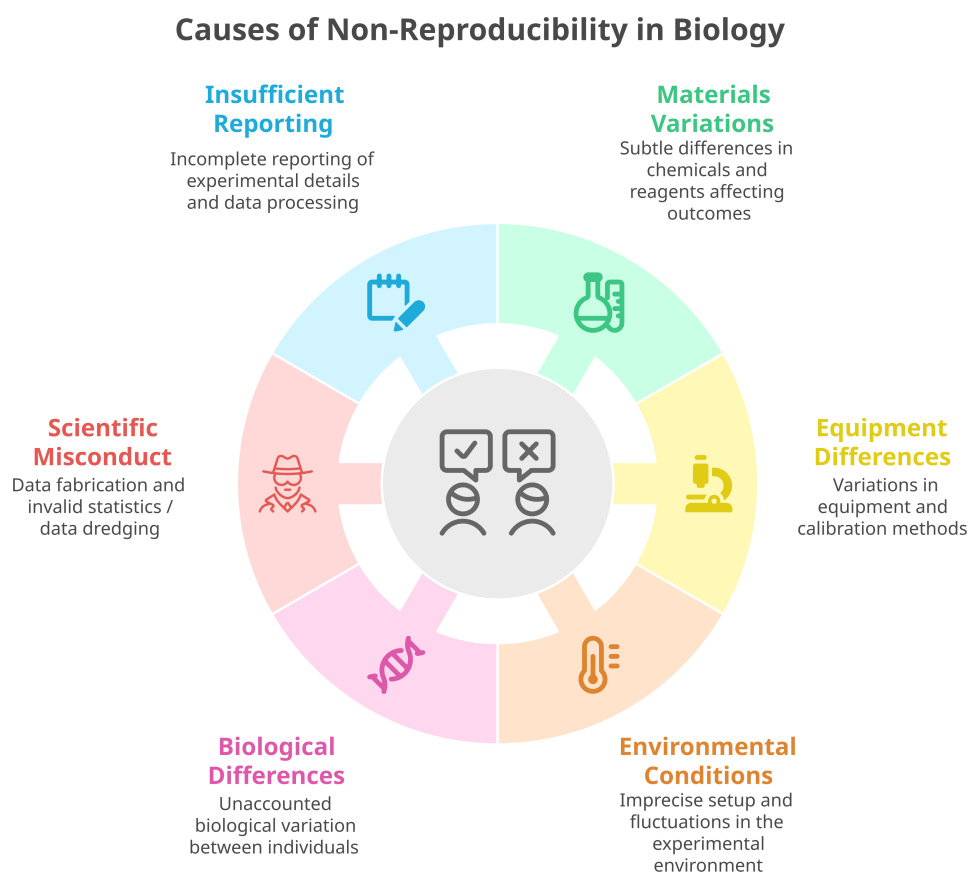
Naturally, there are also less ill-intended causal sequences involving, e.g. poor experimentation or insufficient in-house validation of experiment outcomes [7,10]. Thus, although the root cause of non-reproducibility may lie elsewhere, incomplete reporting of experimental details, acquired data and analytical processes can be identified as a common symptom of irreproducible research. Additionally, discipline-specific problems arising from common practices that are not critically evaluated have been reported, which already require discipline-specific solutions at the experimental stage prior to publication [10,14,15]. Across disciplines, however, enforcing complete reporting appears to be the best approach to boost reproducibility while simultaneously filtering out bad and fraudulent science.

Guidelines for research publications have been put in place to further good scientific publication practice and, consequently, mitigate non-reproducibility. A major set of guidelines for animal experimental research was the publication of the ARRIVE (Animal Research: Reporting of In Vivo Experiments) guidelines in 2010 [16]. The ARRIVE guidelines aimed to set a reporting standard for animal research and defined recommendations for aspects to be included in research papers. ARRIVE not only provided recommendations for the structure and content of research publications but also defined experimental aspects that need to be reported and how these should best be reported. The ARRIVE guidelines received an extensive update in 2020 with ARRIVE 2.0 and the *ARRIVE Essential 10* [17,18]. Additionally, funding agencies such as the German Research Foundation (DFG) have recognized the need and published guidelines on good scientific practice as binding requirements for institutions funded by these agencies. Recently, the Materials Design Analysis Reporting (MDAR) initiative published the MDAR framework for life sciences, including a checklist for transparent reporting [19]. The MDAR framework aims to set minimum standards and is designed to be enhanced by more specialized guidelines, such as ARRIVE for animal experimentation [19] or even more specialized guidelines for specific methods [20,21].

Currently, many publishers have adopted guidelines such as MDAR or ARRIVE, the real-life impact of which, however, has yet to be shown. Despite these guidelines, the reporting of experimental details is often incomplete. An investigation of 238 biomedical research articles found that only 54% of the resources (e.g. antibodies and cell lines) used in these studies were identifiable [2]. An analysis of biomedical research papers involving imaging showed that, on average, only 138 words were devoted to describing the details of image acquisition, which is insufficient to describe all the necessary details [14]. A survey of studies performing respirometry in aquatic animals found that, for the 202 papers analysed, critical aspects were missing in almost half of the publications, although with a positive trend since 1993 [20]. A recent analysis of 84 research articles on organismal animal physiology found that approximately one-third of the experimental details applicable to the respective studies had not been reported [8]. It is striking that some aspects of animal experimentation, despite their known impact, have been reported in less than 25% of the analysed publications [8]. One of these under-reported variables was the time of day at which the experiment was performed, which is known to affect an organism's physiology based on circadian rhythms [22]. From such analyses of experimental reporting, it is plausible to conclude that incomplete reporting of materials and methods may significantly contribute to non-reproducibility [8,14].

## 2. Implications for publishers, editors and reviewers

Enforcement of existing standards must be increased. The roles and responsibilities need to be clear: reviewers need to thoroughly review papers and should be encouraged to spend more time checking whether experimental and analytical details have been reported in sufficient detail. Editors should not rely solely on reviewers; checklists based on guidelines such as MDAR or ARRIVE, which may already be implemented, need to be put into practice. This means that editors and reviewers must be able to make a sufficiently reliable assessment of the reporting completeness. This extra effort must also be recognized by publishers by valuing thoroughness of reviews over speed. However, an aspect that might easily go unnoticed is the choice of reviewers. It is essential that reviewers are entirely independent from the authors of the manuscript they are assessing. While I have just started as an associate editor, I was unpleasantly surprised by the corresponding authors, suggesting as potential reviewers their own (former) PhD students and close collaborators with several joint publications within the last 5 years. Finding such connections is crucial to guarantee impartial reviews, but requires extra effort. Artificial intelligence (AI)-assisted searches of publication databases might be implemented in online submission systems and provide information such as the number of joint publications of the authors and the suggested reviewers. In addition, journals can create incentives for reviewers to engage in thorough review. Many journals allow editors to rate review reports to enable editors to select reliable reviewers based on their high ratings. To motivate reviewers to achieve high ratings through good reports, publishers might give discounts on article processing charges to researchers that contribute to their journals' success by performing thorough reviews. To facilitate the assessment of completeness of the reporting of experimental details for the reviewers, AI-driven text analysis might extract the provided experimental information for easier judgement of whether the given information is sufficient to allow for reproducibility.

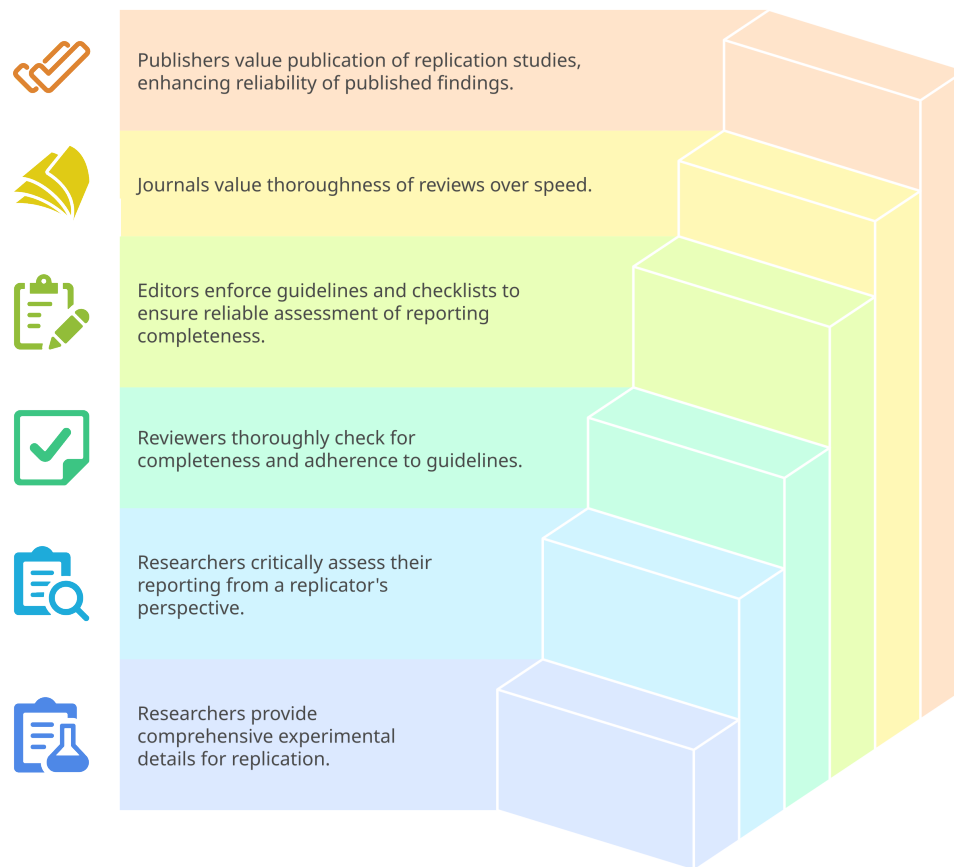


**Figure 1.** Some of the main factors leading to non-reproducibility in experimental biology.

Furthermore, journals and publishers must increase the value of replication studies, that is, the repetition of entire research studies or parts thereof. Replications are intended to validate or falsify the outcomes of a previous study and allow for assumptions on the reproducibility and generalizability of the results of the original study [23]. If a replication study is unlikely to be published, there is little incentive for researchers to spend time and resources on it. Thus, the non-reproducibility of a study remains undiscovered until, for instance, research attempts to build upon the said study and fails to do so, only to unveil non-reproducibility later. *The Royal Society* made a step in the right direction in 2018 by introducing the initiative for replication studies in psychology and cognitive neuroscience, developing an entire workflow to allow for fair and outcome-neutral peer review of replication studies. While some other journals, such as *Pharmacology Research & Perspectives* or *Microbiology Spectrum*, also explicitly support the publication of replication studies, other publishers and journals have yet to follow in order to provide replication studies, that is, the empirical data basis for all reproducibility assessments, their rightful place. Journals appear to be reluctant to publish replication studies, because innovative papers are the main drivers of a journal's impact factor, even if they turn out to be flawed [24,25]. The scientific value of replication studies, both validating and falsifying, is not adequately represented by their low ability to publish such studies, as has been elaborated before [23,24]. Possible solutions could include openly available online sections or journals devoted to the publication of replication studies. Such separation should be considered a valuable investment by publishers in the reliability and trustworthiness of the findings published in their flagship journals while being neutral regarding the journals' impact factor. Currently, an international initiative led by a team from the University of Münster (Germany), funded by the federal state of North Rhine-Westphalia, prepares to launch *Replication Research* (R2, ISSN: 3052-5977), a Diamond Open Access (free to read and to publish) journal devoted to publishing replication studies focusing on reproducibility and comprehensive reporting for an increasing number of scientific disciplines [26]. To provide resources to engage in replication studies, the Dutch Research Council (NWO) continues its national programme Open Science NL, which targets the replication of landmark studies [27]. These are only a few examples that may inspire similar efforts.

### 3. Implications for authors/researchers

The first and most important step towards reproducibility is as difficult as it is obvious: every detail necessary for successful reproduction needs to be reported. While such cases may exist, it seems more realistic to assume that only a minority of studies intentionally leave out crucial details to prevent replication studies which might expose weaknesses in the researchers' experimental design. Therefore, unawareness of the impact of certain aspects of experimentation is an obstacle that needs to be overcome. It is inevitable to place ourselves in the position of potential replicators or reviewers and critically evaluate the completeness of our own reporting. This includes reporting the small but critical procedural adjustments that researchers often



**Figure 2.** Steps to enhance reproducibility in scientific publications—a shared effort between all entities involved in the process of scientific publication.

discover through experience. Such undocumented optimizations are commonly noted informally in laboratory notebooks and can include minor tweaks which make devices perform as desired or a certain way of handling that prevents samples from sticking to the wall of a dish. Because detailed descriptions may simply be too long for a paper to be readable, a dual strategy could be established, where a brief description of the applied methods is part of the main text, while the complete description is provided as a supplement. Whether we may be reluctant to share such details or simply unaware that they could be of any significance regarding reproducibility, allowing for our experiments to be replicated in the first place by providing all necessary details should be considered a quality feature of a research article.

To this end, we can distinguish reproducibility (of a study) from actual reproduction (of the results) in a manner similar to Karl Popper's (1902–1994) distinction between falsifiability and falsification. Falsifiability describes the general property that a scientific hypothesis can be falsified by empirical methods, whereas falsification describes that empirical evidence indeed proves the hypothesis to be wrong [28]. Analogously, reproducibility may describe whether a replication study can be conducted with the given information independent of the outcome. At its core, this is independent of the successful or unsuccessful replication of the original results. It may be beyond our influence whether other researchers can reproduce our results. Therefore, we can only allow for reproducibility of our experiments by providing transparent and complete reporting of our methods and resources. Consequently, unsuccessful replication of results does not automatically render a study non-reproducible. After all, replicators are not infallible, as has recently been shown in a discussion of a protocol using tartrazine to achieve optical clearing in live mice [29]. A failed replication led to a preprint calling the conclusions by Ou *et al* into question [30], but it was later withdrawn, as it became clear that the replicators' protocol differed from the original protocol in crucial steps, and other independent groups managed to apply the original methods successfully [31]. Ultimately, an unsuccessful replication may provide new insights, for instance, regarding which steps are essential for the outcome and, thus, must not be altered.

While the above-mentioned pleas for more rigorous documentation and reporting of experimental methods, the scientific community is increasingly paying attention to possibilities of enhancing reproducibility already at the methodological level prior to publication. One emerging view is that 'heterogenization,' which introduces controlled variability in experimental designs, may be a promising strategy for enhancing reproducibility [32–35]. Paradoxically, increased biological variability can lead to decreased inter-experimental variability, since traditional standardization not only fails to account for biological and environmental diversity but also amplifies subtle differences between laboratories, leading to irreproducible findings [6,33,36]. Meta-analytic evidence demonstrates that embracing variability improves both the reproducibility and generalizability of preclinical results, suggesting that systematic heterogenization could mitigate the 'reproducibility crisis' by making findings more robust across diverse populations and settings [32,33,37]. Interestingly, the heterogenization of experimenters conducting

the investigation has been shown to increase the reproducibility [38,39]. In this special issue, examples of the benefits of introducing controlled variability into the experimental design will be found.

While heterogenization may mark a paradigm shift in biological experiment design, there are also new community-driven ideas on how reproducibility can be increased at the level of data acquisition and analysis. This is exemplified by the efforts of the microscopy and imaging community on openness and transparency. Although the benefits of modern biological imaging are obvious, there is often a lack of transparency regarding the acquisition, processing and analysis of images [14,15]. Furthermore, with more advanced imaging, the outcome increasingly relies on the exact equipment used, which increases the impact of access as a factor for (non-)reproducibility [40]. Transparency in reporting details on image acquisition, as well as technical details of the imaging set-up, are two of the main factors that contribute to the non-reproducibility of bioimaging. The first aspect can be met with guidelines and best practices for documenting and reporting imaging modalities [15,41]. However, allowing for increased reproducibility at the methodological level requires more innovative approaches. The microscopy community, therefore, pushes open science approaches and multidisciplinary to enhance the reproducibility of imaging systems and thus imaging experiments [40,42–46]. Such approaches increase two complementary aspects of reproducibility: increased access allows direct replication of experiments, for example, for hypothesis testing. However, configurable microscopes produce reproducible experimental set-ups for studies of different organisms and allow for better comparability [40].

A multifaceted approach is essential to address non-reproducibility in experimental biology and beyond (figure 2). This includes stricter enforcement of reporting guidelines, increased recognition of replication studies and comprehensive documentation of the experimental details. All instances involved in the process of scientific research and publication should prioritize transparency and thorough reporting, while embracing innovative strategies such as experimental heterogenization and open science approaches. Fostering a culture that values reproducibility and replication will enhance the reliability and impact of biological research and ultimately lead to more robust scientific progress.

**Ethics.** This work did not require ethical approval from a human subject or animal welfare committee.

**Data accessibility.** This article has no additional data.

**Declaration of AI use.** The figures were prepared using Napkin (www.napkin.ai). Paperpal was used for minor language editing.

**Authors' contributions.** T.G.: writing—original draft.

**Conflict of interest declaration.** I declare I have no competing interests.

**Funding.** This work was supported by the German Research Foundation (DFG; grant no. GO 3341/1-1). My current position has been provided by the Alexander von Humboldt Foundation in the framework of the Alexander von Humboldt Professorship to J. Huisken, endowed by the German Federal Ministry of Education and Research.

**Acknowledgements.** I cordially thank Gail Schwieterman, Emily Hardison, Chris Wood and Warren Burggren for inviting me to speak at the symposium 'Embracing Variability in Comparative Physiology: why it matters and what to do with it' at the SEB Conference 2024. I am grateful to Warren Burggren for numerous intense discussions on variability and reproducibility. I gratefully acknowledge the comments and suggestions of the three anonymous reviewers, which were incredibly helpful in improving this manuscript.

## References

1. Freedman LP, Inglese J. 2014 The increasing urgency for standards in basic biologic research. *Cancer Res.* **74**, 4024–4029. (doi:10.1158/0008-5472.CAN-14-0925)
2. Vasilevsky NA, Brush MH, Paddock H, Ponting L, Tripathy SJ, Larocca GM, Haendel MA. 2013 On the reproducibility of science: unique identification of research resources in the biomedical literature. *PeerJ* **1**, e148. (doi:10.7717/peerj.148)
3. Kern P, Cramp RL, Gordos MA, Watson JR, Franklin CE. 2018 Measuring  $U_{crit}$  and endurance: equipment choice influences estimates of fish swimming performance. *J. Fish Biol.* **92**, 237–247. (doi:10.1111/jfb.13514)
4. Bucking C, Bury NR, Sundh H, Wood CM. 2024 Making in vitro conditions more reflective of in vivo conditions for research on the teleost gastrointestinal tract. *J. Exp. Biol.* **227**, jeb246440. (doi:10.1242/jeb.246440)
5. Lucanic M *et al.* 2017 Impact of genetic background and experimental reproducibility on identifying chemical compounds with robust longevity effects. *Nat. Commun.* **8**, 14256. (doi:10.1038/ncomms14256)
6. Crabbe JC, Wahlsten D, Dudek BC. 1999 Genetics of mouse behavior: interactions with laboratory environment. *Science* **284**, 1670–1672. (doi:10.1126/science.284.5420.1670)
7. Baker M. 2016 1,500 scientists lift the lid on reproducibility. *Nature* **533**, 452–454. (doi:10.1038/533452a)
8. Göpel T, Burggren WW. 2022 Insufficient reporting of experimental variables as a cause for nonreproducibility in animal physiology? A case study. *Am. J. Physiol. Regul. Integr. Comp. Physiol.* **323**, R363–R374. (doi:10.1152/ajpregu.00026.2022)
9. Clark TD, Binning SA, Raby GD, Speers-Roesch B, Sundin J, Jutfelt F, Roche DG. 2016 Scientific misconduct: the elephant in the lab. A response to Parker *et al.* *Trends Ecol. Evol.* **31**, 899–900. (doi:10.1016/j.tree.2016.09.006)
10. Parker TH, Forstmeier W, Koricheva J, Fidler F, Hadfield JD, Chee YE, Kelly CD, Gurevitch J, Nakagawa S. 2016 Transparency in ecology and evolution: real problems, real solutions. *Trends Ecol. Evol.* **31**, 711–719. (doi:10.1016/j.tree.2016.07.002)
11. Freedman LP, Cockburn IM, Simcoe TS. 2015 The economics of reproducibility in preclinical research. *PLoS Biol.* **13**, e1002165. (doi:10.1371/journal.pbio.1002165)
12. Smaldino PE, McElreath R. 2016 The natural selection of bad science. *R. Soc. Open Sci.* **3**, 160384. (doi:10.1098/rsos.160384)
13. Brischoux F, Angelier F. 2015 Academia's never-ending selection for productivity. *Scientometrics* **103**, 333–336. (doi:10.1007/s11192-015-1534-5)
14. Marqués G, Pengo T, Sanders MA. 2020 Imaging methods are vastly underreported in biomedical research. *eLife* **9**, e55133. (doi:10.7554/eLife.55133)
15. Montero Llopis P *et al.* 2021 Best practices and tools for reporting reproducible fluorescence microscopy methods. *Nat. Methods* **18**, 1463–1476. (doi:10.1038/s41592-021-01156-w)
16. Kilkenny C, Browne WJ, Cuthill IC, Emerson M, Altman DG. 2010 The ARRIVE guidelines animal research: reporting *in vivo* experiments. *PLoS Biol.* **8**, e1000412. (doi:10.1371/journal.pbio.1000412)

17. Percie du Sert N *et al.* 2020 The ARRIVE guidelines 2.0: updated guidelines for reporting animal research. *J. Cereb. Blood Flow Metab.* **40**, 1769–1777. (doi:10.1177/0271678X20943823)
18. Percie du Sert N *et al.* 2020 Reporting animal research: explanation and elaboration for the ARRIVE guidelines 2.0. *PLoS Biol.* **18**, e3000411. (doi:10.1371/journal.pbio.3000411)
19. Macleod M, Collings AM, Graf C, Kiermer V, Mellor D, Swaminathan S, Sweet D, Vinson V. 2021 The MDAR (materials design analysis reporting) framework for transparent reporting in the life sciences. *Proc. Natl Acad. Sci. USA* **118**, e2103238118. (doi:10.1073/pnas.2103238118)
20. Killen SS *et al.* 2021 Guidelines for reporting methods to estimate metabolic rates by aquatic intermittent-flow respirometry. *J. Exp. Biol.* **224**, jeb242522. (doi:10.1242/jeb.242522)
21. Cowger W *et al.* 2020 Reporting guidelines to increase the reproducibility and comparability of research on microplastics. *Appl. Spectrosc.* **74**, 1066–1077. (doi:10.1177/0003702820930292)
22. Buijs RM, Soto Tinoco EC, Hurtado Alvarado G, Escobar C. 2021 Chapter 15 – the circadian system: from clocks to physiology. In *Handbook of clinical neurology*, vol. 179 (eds DF Swaab, F Kreier, PJ Lucassen, A Salehi, RM Buijs), pp. 233–247. Amsterdam, The Netherlands: Elsevier.
23. Vachon B *et al.* 2021 Changing research culture toward more use of replication research: a narrative review of barriers and strategies. *J. Clin. Epidemiol.* **129**, 21–30. (doi:10.1016/j.jclinepi.2020.09.027)
24. Curtis MJ, Abernethy DR. 2015 Replication - why we need to publish our findings. *Pharmacol. Res. Perspect.* **3**, e00164. (doi:10.1002/prp2.164)
25. Serra-Garcia M, Gneezy U. 2021 Nonreplicable publications are cited more than replicable ones. *Sci. Adv.* **7**, eabd1705. (doi:10.1126/sciadv.abd1705)
26. Röseler L *et al.* 2025 A Community Model for Rigorous and Inclusive Scholarship: Inaugural Editorial of Replication Research (R2). *Replication Research* (doi:10.17879/replicationresearch-2025-9022)
27. Dutch Research Council (NWO) 2025 New call: replication research becomes more visible and normalised. See <https://www.openscience.nl/en/news/new-call-replication-research-becomes-more-visible-and-normalised> (accessed 27 August 2025).
28. Popper K. 1935 *Logik der Forschung: Zur Erkenntnistheorie der moderner Naturwissenschaft.* (eds P Frank, M Schlick). Vienna, Austria: Julius Springer.
29. Ou Z *et al.* 2024 Achieving optical transparency in live animals with absorbing molecules. *Science* **385**, eadm6869. (doi:10.1126/science.adm6869)
30. Inagaki S, Imai T. 2024 Tartrazine cannot make live tissues transparent. *bioRxiv*.
31. Narawane A, Trout R, Viehland C, Kuo A, Vajzovic L, Dhalla AH, Toth CA. 2024 Optical clearing with tartrazine enables deep transscleral imaging with optical coherence tomography. *J. Biomed. Opt.* **29**, 120501. (doi:10.1117/1.JBO.29.12.120501)
32. Richter SH, Garner JP, Würbel H. 2009 Environmental standardization: cure or cause of poor reproducibility in animal experiments? *Nat. Methods* **6**, 257–261. (doi:10.1038/nmeth.1312)
33. Richter SH, Garner JP, Auer C, Kunert J, Würbel H. 2010 Systematic variation improves reproducibility of animal experiments. *Nat. Methods* **7**, 167–168. (doi:10.1038/nmeth0310-167)
34. Karp NA. 2018 Reproducible preclinical research—is embracing variability the answer? *PLoS Biol.* **16**, e2005413. (doi:10.1371/journal.pbio.2005413)
35. Voelkl B *et al.* 2020 Reproducibility of animal research in light of biological variation. *Nat. Rev. Neurosci.* **21**, 384–393. (doi:10.1038/s41583-020-0313-3)
36. Voelkl B, Würbel H. 2016 Reproducibility crisis: are we ignoring reaction norms? *Trends Pharmacol. Sci.* **37**, 509–510. (doi:10.1016/j.tips.2016.05.003)
37. Usui T, Macleod MR, McCann SK, Senior AM, Nakagawa S. 2021 Meta-analysis of variation suggests that embracing variability improves both replicability and generalizability in preclinical research. *PLoS Biol.* **19**, e3001009. (doi:10.1371/journal.pbio.3001009)
38. von Kortzfleisch VT *et al.* 2022 Do multiple experimenters improve the reproducibility of animal studies? *PLoS Biol.* **20**, e3001564. (doi:10.1371/journal.pbio.3001564)
39. Nigri M, Åhlgren J, Wolfer DP, Voikar V. 2022 Role of environment and experimenter in reproducibility of behavioral studies with laboratory mice. *Front. Behav. Neurosci.* **16**, 835444. (doi:10.3389/fnbeh.2022.835444)
40. Power RM, Huiskens J. 2019 Putting advanced microscopy in the hands of biologists. *Nat. Methods* **16**, 1069–1073. (doi:10.1038/s41592-019-0618-1)
41. Schmied C *et al.* 2024 Community-developed checklists for publishing images and image analyses. *Nat. Methods* **21**, 170–181. (doi:10.1038/s41592-023-01987-9)
42. Pitrone PG, Schindelin J, Stuyvenberg L, Preibisch S, Weber M, Eliceiri KW, Huiskens J, Tomancak P. 2013 OpenSPIM: an open-access light-sheet microscopy platform. *Nat. Methods* **10**, 598–599. (doi:10.1038/nmeth.2507)
43. Weber M, Huiskens J. 2021 Multidisciplinarity is critical to unlock the full potential of modern light microscopy. *Front. Cell Dev. Biol.* **9**, 739015. (doi:10.3389/fcell.2021.739015)
44. Diederich B, Lachmann R, Carlstedt S, Marsikova B, Wang H, Uwurukundo X, Mosig AS, Heintzmann R. 2020 A versatile and customizable low-cost 3D-printed open standard for microscopic imaging. *Nat. Commun.* **11**, 5979. (doi:10.1038/s41467-020-19447-9)
45. Diederich B, Müllenbroich C, Vladimirov N, Bowman R, Stirling J, Reynaud EG, Andreev A. 2022 CAD we share? Publishing reproducible microscope hardware. *Nat. Methods* **19**, 1026–1030. (doi:10.1038/s41592-022-01484-5)
46. Wenzel T. 2023 Open hardware: from DIY trend to global transformation in access to laboratory equipment. *PLoS Biol.* **21**, e3001931. (doi:10.1371/journal.pbio.3001931)